

MS:

$$7.3 \quad (a) \int (4x + 3) dx = 2x^2 + 3x + c$$

$$(c) \int (-2x + 3 - 4x^3) dx = -x^2 + 3x - x^4 + c$$

$$(k) \int \left(\frac{1}{x} + 3x^2\right) dx = \ln|x| + x^3 + c$$

$$(m) \int 2e^{5x} dx = \frac{2}{5}e^{5x} + c$$

$$(p) \int \ln(x^2) dx$$

$$f = \ln x^2, \quad f' = \frac{2}{x}, \quad g = x, \quad g' = 1$$

$$\int \ln x^2 = x \ln x^2 - \int x \cdot \frac{2}{x} dx = x \ln x^2 - \int 2 dx = x \ln(x^2) - 2x + c$$

$$7.6 \quad (c) \int_1^{16} (5x^{3/2} - 2x^{-5/4}) dx$$

$$= 2x^{5/2} + 8x^{-1/4} \Big|_1^{16}$$

$$= \left(2(16)^{5/2} + 8(16)^{-1/4}\right) - (2 + 8) = (2048 + 4) - 10 = 2042$$

$$(i) \int_2^4 (3x^2 + x + 5) dx$$

$$= \left[x^3 + \frac{x^2}{2} + 5x\right]_2^4 = 72$$

$$(j) \int_2^2 (3x^2 + x + 5) dx$$

$$= \left[x^3 + \frac{x^2}{2} + 5x\right]_2^2 = 0$$

$$(k) \int_2^4 (3x^4 + 2x^3 + x^2 + x + 1) dx + \int_4^6 (3x^4 + 2x^3 + x^2 + x + 1) dx$$

$$= \int_2^6 (3x^4 + 2x^3 + x^2 + x + 1) dx = \left[\frac{3x^5}{5} + \frac{x^4}{2} + \frac{x^3}{3} + \frac{x^2}{2} + x\right]_2^6$$

$$= \left(\frac{23328}{5} + 648 + 72 + 18 + 6\right) - \left(\frac{96}{5} + 8 + \frac{8}{3} + 2 + 2\right) = \frac{80636}{15}$$

$$(l) \int_2^6 (3x^4 + 2x^3 + x^2 + x + 1) dx = \frac{80636}{15}$$

Same value as (k): the two pieces over $[2, 4]$ and $[4, 6]$ add up to the integral over $[2, 6]$.

From Sydsaeter and Hammond

$$9.3.7 \quad (a) \frac{d}{dt} \int_0^t x^2 dx = \frac{d}{dt} \left[\frac{x^3}{3}\right]_0^t = t^2$$

$$(b) \frac{d}{dt} \int_t^3 e^{-x^2} dx = \boxed{-e^{-t^2}}$$

$$9.5.3 \quad (a) \int_1^4 \sqrt{t} \ln t \, dt$$

$$f = \ln t, \quad g = \frac{2}{3} t^{3/2}$$

$$f' = \frac{1}{t}, \quad g' = \sqrt{t}$$

$$= \frac{2}{3} t^{3/2} \ln(t) \Big|_1^4 - \frac{2}{3} \int_1^4 t^{1/2} dt$$

$$= \frac{2}{9} t^{3/2} (3 \ln(t) - 2) \Big|_1^4$$

$$= \boxed{\frac{16}{9} [3 \ln(4) - 2] + \frac{4}{9}}$$

$$(b) \int_0^2 (x-2)e^{-x/2} dx$$

$$f = x-2, \quad f' = 1, \quad g = -2e^{-x/2}, \quad g' = e^{-x/2}$$

$$= -2(x-2)e^{-x/2} \Big|_0^2 + 2 \int_0^2 e^{-x/2} dx$$

$$= -2(x-2)e^{-x/2} + 2(-2e^{-x/2}) \Big|_0^2$$

$$= e^{-x/2} [-2x + 4 - 4]_0^2 = [-2xe^{-x/2}]_0^2 = \boxed{\frac{-4}{e}}$$

$$(c) \int_0^3 (3-x)e^x dx$$

$$f = (3-x), \quad f' = -1, \quad g = e^x, \quad g' = e^x$$

$$= (3-x)e^x + \int e^x dx$$

$$= e^x [3-x+1] = e^x (4-x) \Big|_0^3$$

$$= \boxed{e^3 - 4}$$

Additional Question:

The equilibrium quantity is determined by the equation $50 - 0.1Q^* = 0.2Q^* + 20$, which gives $Q^* = 100$. Then $P^* = 0.2Q^* + 20 = 40$. Hence,

$$CS = \int_0^{100} [50 - 0.1Q - 40] dQ = \int_0^{100} [10 - 0.1Q] dQ = [10Q - 0.05Q^2]_0^{100} = \boxed{500}$$

$$PS = \int_0^{100} [40 - (0.2Q + 20)] dQ = \int_0^{100} [20 - 0.2Q] dQ = [20Q - 0.1Q^2]_0^{100} = \boxed{1000}$$

The End.